

THE DEMONSTRATIONS, TO BE READ ALOUD

Theorems 1–37

Augros, Introduction to Arithmetic and Geometry, Chapter 1

One page for each theorem. Every letter and every numbered angle follows the figures in the book, so this can be read straight through while you point at the drawing or build it on the board.

THEOREM 1

How to make an equilateral triangle A CONSTRUCTION

Given the straight line AB.

We are to make an equilateral triangle on it, with AB as its base.

About the point A, with radius AB, draw the circle X — Postulate 3.

About the point B, with radius BA, draw the circle Z — Postulate 3.

These two circles cut each other. Call the crossing point above the line C.

Join A to C — Postulate 1.

Join B to C — Postulate 1.

Now, AC equals AB, because both are radii of circle X — Definition 17.

And BC equals AB, because both are radii of circle Z — Definition 17.

So AC equals BC, because each of them is equal to AB — Common Notion 1.

All three sides of triangle ABC are equal to one another.

Therefore triangle ABC is equilateral — Definition 20 — and it stands on AB as its base.

That is what we set out to make. **Q.E.F.**

THEOREM 2

Side-Angle-Side A THEOREM

Here are two triangles, ABC and DEF.

In them, AB equals DE; the angle ABC equals the angle DEF; and BC equals EF.

We are to prove that everything else about the two triangles is equal as well.

Pick up triangle ABC in your imagination, and lay it down on triangle DEF, so that the side AB falls on the side DE.

Since AB equals DE, the two lines coincide exactly: A sits on D, and B sits on E.

Since the angle ABC equals the angle DEF, the side BC falls along the side EF.

And since BC equals EF, the point C sits exactly on the point F.

So A is on D, B is on E, and C is on F.

Then the side AC must fall exactly along the side DF.

For if it fell outside, the two straight lines AC and DF would cut each other twice — at A and again at C — and enclose a space between them. Two straight lines cannot do that.

So all three sides coincide, and the two triangles coincide altogether.

Things that can be made to coincide are equal — Common Notion 4.

Therefore AC equals DF; the angle BAC equals the angle EDF; the angle BCA equals the angle EFD; and the two triangles have equal areas. **Q.E.D.**

THEOREM 3

Base angles of an isosceles triangle A THEOREM

Triangle ABC is isosceles: AB equals AC — Definition 20.

We are to prove two things: that the base angles ABC and ACB are equal, and that the angles under the base, DBC and ECB, are equal.

Extend AB beyond B to any point D — Postulate 2.

Extend AC beyond C — Postulate 2 — and about C, with radius BD, draw a circle cutting that extension at E, so that CE equals BD — Postulate 3.

Join B to E, and join C to D — Postulate 1.

Look first at triangle ADC and triangle AEB.

AB equals AC, which is given; and BD equals CE, by the construction.

Add equals to equals: AD equals AE — Common Notion 2.

The angle at A belongs to both triangles alike.

So in triangle ADC and triangle AEB we have a side, the angle between, and the next side, equal each to each.

Therefore the two triangles are equal in every corresponding part — Side-Angle-Side, Theorem 2.

From that, the angle ADC equals the angle AEB, and DC equals EB.

Now look at triangle BDC and triangle CEB.

BD equals CE, by the construction; DC equals EB, just proved; and the angle BDC equals the angle CEB, just proved.

Again a side, the angle between, and the next side are equal each to each.

Therefore these two triangles also are equal in every corresponding part — Side-Angle-Side, Theorem 2.

From that, the angle DBC equals the angle ECB. Those are the angles under the base, and that is the second thing we wanted.

Now take the whole angles. The angle ABE equals the angle ACD, from the first pair of triangles.

And the angle EBC equals the angle DCB, from the second pair.

Take equals away from equals, and the remainders are equal — Common Notion 3.

Therefore the angle ABC equals the angle ACB.

The base angles of an isosceles triangle are equal. **Q.E.D.**

THEOREM 4

Converse of the isosceles base-angle theorem A THEOREM

In triangle ABC the angle ABC equals the angle ACB.

We are to prove that AB equals AC — that the triangle is isosceles.

Suppose it were not so. Then one of the two sides would be the greater. Let AB be the greater.

From AB cut off BD equal to AC, by drawing a circle about B with radius AC — Postulate 3.

Join C to D — Postulate 1.

Now look at triangle DBC and triangle ACB.

BD equals AC, by the construction.

The angle DBC equals the angle ACB, which is what we were given.

And the side BC belongs to both triangles alike.

So a side, the angle between, and the next side are equal, each to each.

Therefore the two triangles are equal in every corresponding part, and in particular they are equal in area — Side-Angle-Side, Theorem 2.

But triangle DBC is only a part of triangle ACB, and every whole is greater than any one of its parts — Common Notion 5.

So the part would have to equal the whole. That cannot be.

Therefore AB is not greater than AC.

In exactly the same way, AC is not greater than AB.

Neither side is greater than the other. Therefore AB equals AC. **Q.E.D.**

THEOREM 5

The three sides determine the angles — triangles are rigid A THEOREM

A square made of four hinged boards can be pushed over: it leans into a new shape, with all four sides still the same length.

We ask whether a triangle can do the same — keep the lengths of its three sides, and change its angles.

Suppose it could. Then there would be two different triangles, ABM and ABN, standing on the same base AB, with AM equal to AN and BM equal to BN, but with different angles — so that M and N are two different points.

Join M to N — Postulate 1.

In triangle AMN, AM equals AN. It is isosceles, so its base angles are equal: the angle AMN equals the angle ANM — Theorem 3.

The angle AMN is the angle marked 1 together with the angle marked 2, and the angle ANM is the angle marked 3.

So angle 1 plus angle 2 equals angle 3.

In triangle BMN, BM equals BN. It too is isosceles, so its base angles are equal: the angle BMN equals the angle BNM — Theorem 3.

The angle BMN is the angle marked 2, and the angle BNM is the angle marked 3 together with the angle marked 4.

So angle 2 equals angle 3 plus angle 4.

Now put the second result into the first. In place of angle 2, write angle 3 plus angle 4.

Then angle 1, plus angle 3, plus angle 4, equals angle 3.

That says an angle is equal to itself with two more angles added on — that a whole equals one of its own parts. But every whole is greater than any one of its parts — Common Notion 5.

That cannot be.

Therefore there is no second triangle.

In a triangle, the lengths of the three sides determine the three angles: you cannot change any angle while keeping all three side lengths the same. This is what is meant by saying that triangles are rigid. **Q.E.D.**

THEOREM 6

Side-Side-Side A THEOREM

Here are two triangles whose three sides are equal, each to each.

We are to prove that their corresponding angles are equal too, and that they have equal areas.

Lay them on the same base CD , with the equal sides at the same ends, so that CA equals CB and DA equals DB . Let A be the top of the one, and B the top of the other.

Suppose A and B were two different points.

Then we would have two different triangles standing on the same base CD , with all three sides of the one equal to all three sides of the other, and yet slanting differently — that is, with different angles.

But a triangle cannot do that: in a triangle the lengths of the three sides determine the three angles — Theorem 5.

So A and B are not two different points. They are one and the same point.

Then CA lies exactly along CB , and DA lies exactly along DB , and every angle of the one lies exactly on the corresponding angle of the other.

Things that can be made to coincide are equal — Common Notion 4.

Therefore the corresponding angles of the two triangles are equal, and their areas are equal.

Q.E.D.

THEOREM 7

How to bisect any rectilineal angle A CONSTRUCTION

Given the rectilineal angle ABC.

We are to cut it into two equal parts.

Take any point D on the side BA.

About B, with radius BD, draw a circle, cutting the side BC at E — Postulate 3.

Then BE equals BD, since both are radii of that circle — Definition 17.

Join D to E — Postulate 1.

On DE make the equilateral triangle DEF — Theorem 1.

Join B to F — Postulate 1. This line BF is the one we want.

Now look at triangle FBD and triangle FBE.

BD equals BE, by the construction.

FD equals FE, because DEF is equilateral — Definition 20.

And FB belongs to both triangles alike.

So the three sides of the one are equal to the three sides of the other, each to each.

Therefore their corresponding angles are equal — Side-Side-Side, Theorem 6.

Therefore the angle FBD equals the angle FBE.

So the straight line BF has cut the angle ABC into two equal parts. **Q.E.F.**

THEOREM 8

How to bisect any straight line A CONSTRUCTION

Given the straight line AB.

We are to cut it into two equal parts.

On AB make the equilateral triangle ABC — Theorem 1.

Cut the angle ACB into two equal parts with the straight line CD, and let CD meet AB at the point D — Theorem 7.

Now look at triangle ACD and triangle BCD.

AC equals BC, because ABC is equilateral — Definition 20.

The angle ACD equals the angle BCD, because CD bisects the angle at C — that is what Theorem 7 gave us.

And CD belongs to both triangles alike.

So in triangle ACD and triangle BCD we have a side, the angle between, and the next side, equal each to each.

Therefore the two triangles are equal in every corresponding part — Side-Angle-Side, Theorem 2.

Therefore AD equals DB.

So the straight line AB has been cut into two equal parts at the point D. **Q.E.F.**

THEOREM 9

Erect a perpendicular from a point on a line A CONSTRUCTION

Given the straight line AB, and the point P on it.

We are to draw from P a straight line at right angles to AB.

Take any point C on PA.

About P, with radius PC, draw a circle, cutting PB at D — Postulate 3.

Then PD equals PC, since both are radii of that circle — Definition 17.

On CD make the equilateral triangle CDR — Theorem 1.

Join P to R — Postulate 1. This is the line we want.

Now look at triangle RPC and triangle RPD.

PC equals PD, by the construction.

CR equals DR, because CDR is equilateral — Definition 20.

And RP belongs to both triangles alike.

So the three sides of the one are equal to the three sides of the other, each to each.

Therefore their corresponding angles are equal — Side-Side-Side, Theorem 6.

Therefore the angle RPC equals the angle RPD.

Now these two angles are adjacent angles, formed by the one line RP standing on the other line AB, and they are equal to each other.

When one straight line stands on another and makes the adjacent angles equal, each of those angles is a right angle — Definition 9.

So the angle RPC and the angle RPD are right angles, and PR is perpendicular to AB — Definition 10.

So a straight line has been set up at right angles to AB, from the point P on it. **Q.E.F.**

THEOREM 10

Drop a perpendicular from a point to a line A CONSTRUCTION

Given the straight line AB, and the point P above it.

We are to drop from P a straight line perpendicular to AB.

Choose any point D below AB.

About P, with radius PD, draw a circle — Postulate 3. Because D is below the line, this circle cuts AB in two places. Call them G and E.

Join G to P, and join E to P — Postulate 1.

Cut GE into two equal parts at the point H — Theorem 8.

Join P to H — Postulate 1. This is the line we want.

Now look at triangle PHG and triangle PHE.

PG equals PE, because both are radii of the circle about P — Definition 17.

HG equals HE, because we bisected GE at H.

And PH belongs to both triangles alike.

So the three sides of the one are equal to the three sides of the other, each to each.

Therefore their corresponding angles are equal — Side-Side-Side, Theorem 6.

Therefore the angle PHG equals the angle PHE.

But these two equal angles are adjacent angles, formed by the one line PH standing on the other line AB.

Therefore each of them is a right angle — Definition 9.

So PH has been drawn perpendicular to AB — Definition 10. **Q.E.F.**

THEOREM 11

Adjacent angles on a straight line A THEOREM

If PB stands on CD at right angles, the thing is plain: the two adjacent angles are both right angles, and together they make two right angles.

So take the other case. The straight line AB stands on the straight line CD at the point B, but not at right angles.

We are to prove that the angle ABC and the angle ABD together still equal two right angles.

From B draw BP at right angles to CD — Theorem 9.

Then the angle PBC and the angle PBD are both right angles — Definition 9 — so together they make two right angles.

Now look at the pieces. The line BA falls between BP and BD, and cuts the angle PBD in two.

The angle CBP is the angle marked 1; the angle PBA is the angle marked 2; the angle ABD is the angle marked 3.

So the angle PBC together with the angle PBD is the same as angle 1 plus angle 2 plus angle 3.

Therefore angle 1 plus angle 2 plus angle 3 equals two right angles — Common Notion 1.

But angle 1 together with angle 2 is exactly the angle ABC, and angle 3 is the angle ABD.

Therefore the angle ABC together with the angle ABD equals two right angles — Common Notion 1. **Q.E.D.**

THEOREM 12

Two right angles make one straight line A THEOREM

At the point P, the angle APB and the angle BPC together equal two right angles.

We are to prove that AP and PC form one straight line.

Suppose they did not.

Then extend AP straight on through P to some point X — Postulate 2 — where X is not on PC.

Call the angle APB the angle 1, the angle BPX the angle 2, and the angle XPC the angle 3.

Since APX is one straight line, and PB stands on it, angle 1 plus angle 2 equals two right angles — Theorem 11.

But we were given that the angle APB together with the angle BPC equals two right angles; that is, angle 1 plus angle 2 plus angle 3 equals two right angles.

All right angles are equal to one another — Postulate 4 — so these two amounts are one and the same amount.

Therefore angle 1 plus angle 2 equals angle 1 plus angle 2 plus angle 3 — Common Notion 1.

But the first is only a part of the second, and every whole is greater than any one of its parts — Common Notion 5.

That cannot be.

So there is no such point X off the line PC.

Therefore PC is the straight continuation of AP: AP and PC form one straight line. **Q.E.D.**

THEOREM 13

Vertical angles are equal A THEOREM

Two straight lines, AB and CD, cut across each other at the point P.

They make four angles: 1, 2, 3 and 4. The angles marked 1 and 3 lie opposite each other across P, and so do the angles marked 2 and 4. These are the vertical angles.

We are to prove that angle 1 equals angle 3, and that angle 2 equals angle 4.

Angle 1 and angle 2 are adjacent angles where one line stands on a straight line. So together they equal two right angles — Theorem 11.

Angle 3 and angle 2 are also adjacent angles where one line stands on a straight line. So together they equal two right angles — Theorem 11.

Things equal to the same thing are equal to each other. So angle 1 plus angle 2 equals angle 3 plus angle 2 — Common Notion 1.

Take angle 2 away from each side. When equals are taken from equals, the remainders are equal — Common Notion 3.

Therefore angle 1 equals angle 3.

In the same way, angle 2 equals angle 4.

So when two straight lines cut each other, the vertical angles are equal. **Q.E.D.**

THEOREM 14

Exterior angle greater than remote interior A THEOREM

In triangle ABC, the side BC is extended to the point D.

The angle ACD is called the exterior angle.

We are to prove that it is greater than the angle BAC, and greater than the angle ABC.

Cut the side AC into two equal parts at the point E — Theorem 8.

Join B to E — Postulate 1 — and extend BE straight on to the point F, cutting off EF equal to BE — Postulate 2.

Join C to F — Postulate 1.

Now look at triangle AEB and triangle CEF.

EF equals BE, because we made it so.

EA equals EC, because we bisected AC at E.

And the angle AEB equals the angle CEF, because they are vertical angles — Theorem 13.

So a side, the angle between, and the next side are equal, each to each.

Therefore the two triangles are equal in every corresponding part — Side-Angle-Side, Theorem 2.

Therefore the angle marked 1, which is the angle BAC, equals the angle marked 2, which is the angle ECF.

But the angle marked 2 is only a part of the angle ACD, because the line CF falls inside the angle ACD.

Every whole is greater than any one of its parts — Common Notion 5.

So the angle ACD is greater than the angle marked 2, and therefore greater than the angle BAC.

That is the first half of what we wanted.

For the second half, do the same thing on the other side: bisect BC, and extend AC beyond C to a point G.

By the very same argument, the angle BCG is greater than the angle ABC.

But the angle BCG and the angle ACD are vertical angles, and so they are equal — Theorem 13.

Therefore the angle ACD is greater than the angle ABC as well.

So the exterior angle of a triangle is greater than either of the interior and opposite angles.

Q.E.D.

THEOREM 15

Greater side opposite greater angle A THEOREM

In triangle ABC the side AC is greater than the side AB.

We are to prove that the angle opposite AC is greater than the angle opposite AB — that the angle ABC is greater than the angle BCA.

Since AC is greater than AB, we can cut off from AC a part AD equal to AB.

Join B to D — Postulate 1.

Now triangle ABD has AD equal to AB, so it is isosceles.

Its base angles are equal — Theorem 3. So the angle marked 1, which is the angle ABD, equals the angle marked 2, which is the angle ADB.

Next look at triangle BDC.

Its side CD, extended beyond D to A, makes the angle marked 2 the exterior angle of that triangle.

An exterior angle is greater than either of the interior and opposite angles — Theorem 14.

So the angle marked 2 is greater than the angle marked 3, which is the angle BCA.

Angle 1 equals angle 2, and angle 2 is greater than angle 3. So angle 1 is greater than angle 3.

And the whole angle ABC is greater than the angle marked 1, because angle 1 is only a part of it — Common Notion 5.

So the angle ABC is greater than angle 1, and angle 1 is greater than angle 3.

Therefore the angle ABC is greater than the angle BCA. **Q.E.D.**

THEOREM 16

Greater angle opposite greater side A THEOREM

In triangle ABC the angle marked 1, which is the angle at B, is greater than the angle marked 2, which is the angle at C.

We are to prove that the side opposite angle 1 is greater than the side opposite angle 2 — that AC is greater than AB.

There are only three possibilities. Either AC equals AB, or AC is less than AB, or AC is greater than AB. One of the three must be true.

Take the first. Suppose AC equalled AB.

Then the triangle would be isosceles, and the angles opposite those two sides would be equal: angle 1 would equal angle 2 — Theorem 3.

But we were given that angle 1 is greater than angle 2.

So AC cannot equal AB.

Take the second. Suppose AC were less than AB.

Then AB would be the greater side, and in any triangle the greater side has the greater angle opposite it — Theorem 15. So angle 2 would be greater than angle 1.

But we were given the opposite.

So AC cannot be less than AB.

AC is neither equal to AB nor less than AB. Only one possibility is left.

Therefore AC is greater than AB. **Q.E.D.**

THEOREM 17

The triangle inequality A THEOREM

Take any triangle ABC.

We are to prove that any two of its sides added together are greater than the third — for instance, that BA and AC together are greater than BC.

Extend BA beyond A to the point D, and make AD equal to AC — Postulate 2 and Postulate 3.

Join C to D — Postulate 1.

Now triangle ACD has AD equal to AC, so it is isosceles, and its base angles are equal — Theorem 3.

Those base angles are the angle marked 2, which is the angle ACD, and the angle marked 1, which is the angle ADC. So angle 2 equals angle 1.

Look at the whole angle BCD, the angle at C in the large triangle DBC.

The angle marked 2 is only a part of it, since the line CA falls inside it, and every whole is greater than any one of its parts — Common Notion 5.

So the angle BCD is greater than angle 2, and therefore greater than angle 1.

Now look at triangle DBC. In it, the angle BCD is greater than the angle marked 1.

The greater angle has the greater side opposite it — Theorem 16.

The side opposite the angle BCD is BD; the side opposite angle 1 is BC.

Therefore BD is greater than BC.

But BD is BA together with AD, and AD equals AC.

Therefore BA and AC together are greater than BC.

The same argument proves the other two cases.

And it follows that any side is greater than the difference of the other two: take AC away from both sides of that inequality, and BA is greater than BC less AC — Common Notion 3. **Q.E.D.**

THEOREM 18

The shortest path to a line is the perpendicular A THEOREM

Given the straight line AB, and the point P above it.

Let PL be the perpendicular dropped from P to AB — Theorem 10 — and let R be any other point on AB.

We are to prove that PR is greater than PL, so that the perpendicular is the shortest line from P to AB.

Look at triangle PLR.

The angle marked 3, which is the angle PLR, is a right angle, since PL is perpendicular to AB — Definition 9 and Definition 10.

The angle marked 1, on the other side of L, is a right angle too, because PL stands on the straight line AB — Theorem 11.

Now the angle marked 1 is the exterior angle of triangle PLR at the point L.

An exterior angle is greater than either of the interior and opposite angles — Theorem 14.

So angle 1 is greater than the angle marked 2, which is the angle PRL.

And angle 3 equals angle 1, since both are right angles — Postulate 4.

Therefore angle 3 is greater than angle 2.

In triangle PLR the greater angle has the greater side opposite it — Theorem 16.

The side opposite angle 3 is PR; the side opposite angle 2 is PL.

Therefore PR is greater than PL.

And R was any point on AB other than L.

Therefore the perpendicular PL is shorter than any other straight line drawn from P to AB.

Q.E.D.

THEOREM 19

Construct a triangle from three given sides A CONSTRUCTION

Given three straight lines X , Y and Z , such that any two of them together are greater than the third.

We are to make a triangle whose three sides are equal to X , Y and Z .

Draw a straight line, and on it mark the points D and A so that DA equals X ; extend it out to E , as far as you need — Postulate 2.

From A cut off AB equal to Y , and from B cut off BF equal to Z — Postulate 3.

About A , with radius AD , draw a circle — Postulate 3.

About B , with radius BF , draw a circle — Postulate 3.

These two circles must cut each other. Here is why.

AD and BF together are greater than AB , so the two circles reach past each other along AB , and overlap.

BA and AD together are greater than BF , so BD is longer than the radius of circle B , and the point D lies outside circle B .

And AB and BF together are greater than AD , so AF is longer than the radius of circle A , and the point F lies outside circle A .

So the two circles overlap, and yet each one passes outside the other. Therefore they cross.

Call the point where they cross C .

Join A to C , and join B to C — Postulate 1.

Now AC is a radius of circle A , so AC equals AD , which is X — Definition 17.

AB equals Y , by the construction.

And BC is a radius of circle B , so BC equals BF , which is Z — Definition 17.

So triangle ABC has been made, with its three sides equal to X , Y and Z . **Q.E.F.**

THEOREM 20

How to copy an angle A CONSTRUCTION

Given the angle X , and given the straight line PR .

We are to make an angle beginning at the point P , equal to the angle X , and having PR as one of its sides.

Take any point A on one leg of the angle X , and any point B on the other leg. Join A to B — Postulate 1.

Now XA , XB and AB are the three sides of triangle XAB , so any two of them together are greater than the third — Theorem 17.

That means a triangle can be built out of those three lengths — Theorem 19.

Extend RP , and on it cut off PS equal to XA , PT equal to XB , and TV equal to AB — Postulate 3. These are the three lengths we shall need.

Now make the triangle PZT , having PZ equal to XA , PT equal to XB , and ZT equal to AB — Theorem 19.

Look at triangle PZT and triangle XAB .

PZ equals XA ; PT equals XB ; and ZT equals AB .

The three sides of the one are equal to the three sides of the other, each to each.

Therefore their corresponding angles are equal — Side-Side-Side, Theorem 6.

Therefore the angle ZPT equals the angle AXB .

And the angle ZPT begins at P , with the side PT lying along PR .

So we have made at P an angle equal to the given angle X , with PR as one of its sides. **Q.E.F.**

THEOREM 21

Angle-Side-Angle A THEOREM

Here are two triangles, ABC and DEF.

In them, the angle BAC equals the angle EDF; the side AB equals the side DE; and the angle ABC equals the angle DEF.

We are to prove that the rest of their sides and angles are equal as well.

Pick up triangle ABC in your imagination, and lay AB on top of DE.

Since AB equals DE, A sits on D, and B sits on E.

Since the angle BAC equals the angle EDF, the side AC lies along the side DF.

Since the angle ABC equals the angle DEF, the side BC lies along the side EF.

Now C is the point where AC and BC meet, and F is the point where DF and EF meet.

Since AC lies along DF, and BC lies along EF, their meeting points are one and the same: C sits on F.

So A coincides with D, B with E, and C with F.

The three sides of the one coincide with the three sides of the other.

Therefore all their corresponding sides and angles are equal, and their areas are equal — Common Notion 4, and Side-Side-Side, Theorem 6. **Q.E.D.**

THEOREM 22

Angle-Angle-Side A THEOREM

Here are two triangles, ABC and DEF.

In them, the angle BCA equals the angle EFD; the angle BAC equals the angle EDF; and the side AB equals the side DE.

Notice that AB and DE are not the sides between the two equal angles. Each of them lies opposite one of the equal angles.

We are to prove that the rest of their sides and angles are equal.

Pick up triangle ABC and lay AB on top of DE.

Since AB equals DE, A sits on D, and B sits on E.

Since the angle BAC equals the angle EDF, the side AC lies along the side DF.

But is AC the same length as DF?

Suppose it were shorter, so that C landed short of F, at a point on DF. Call that point C-prime.

Then triangle D-C-prime-E is just triangle ACB laid down, so the angle at C-prime equals the angle BCA.

But the angle EFD also equals the angle BCA, since the two original triangles were given that way.

So the angle at C-prime equals the angle EFD.

Now D, C-prime and F lie in one straight line. So the angle at C-prime is an exterior angle of triangle E-C-prime-F, and the angle EFD is one of its interior and opposite angles.

An exterior angle is always greater than an interior and opposite angle — Theorem 14. It can never be equal to one.

That cannot be. So AC is not shorter than DF.

Suppose instead that AC were longer, so that C landed beyond F, on DF extended.

Then by the very same argument an exterior angle would equal an interior and opposite angle, which again cannot be.

So AC is not longer than DF.

AC is neither shorter nor longer. Therefore AC equals DF.

Now AB equals DE, the angle BAC equals the angle EDF, and AC equals DF: a side, the angle between, and the next side.

Therefore all their corresponding sides and angles are equal, and their areas are equal — Side-Angle-Side, Theorem 2. **Q.E.D.**

THEOREM 23

Equal alternate angles make parallels A THEOREM

Here are three straight lines: AB, CD and EF.

EF cuts AB at the point E, and cuts CD at the point F.

The angle marked 1 is the angle AEF, and the angle marked 2 is the angle EFD. These are alternate angles: they lie on opposite sides of the line EF, one at each crossing.

We are given that angle 1 equals angle 2.

We are to prove that AB and CD are parallel.

Suppose they were not parallel.

Then EB and FD, extended far enough, would meet on the right at some point X.

That would make a triangle, EFX.

Now angle 1 is the exterior angle of that triangle at E, since AE is the extension of XE.

And angle 2 is one of the interior and opposite angles of that triangle.

An exterior angle is always greater than an interior and opposite angle — Theorem 14.

But we were given that angle 1 equals angle 2.

That cannot be.

So EB and FD do not meet on that side; and by the same argument they do not meet on the other side either.

Therefore AB and CD never meet, however far they are extended: they are parallel — Definition 22. **Q.E.D.**

THEOREM 24

Supplementary co-interior angles make parallels A THEOREM

Two straight lines, AB and CD, are cut by a third line, EF.

The angle marked 2 and the angle marked 3 are the two interior angles on one side of EF.

We are given that angle 2 and angle 3 together equal two right angles.

We are to prove that AB is parallel to CD.

Angle 2 and the angle marked 1 are adjacent angles, where EF stands on the straight line AB.

So angle 2 plus angle 1 equals two right angles — Theorem 11.

But angle 2 plus angle 3 equals two right angles, which is what we were given.

Therefore angle 2 plus angle 3 equals angle 2 plus angle 1 — Common Notion 1.

Take angle 2 away from each side. When equals are taken from equals, the remainders are equal — Common Notion 3.

Therefore angle 3 equals angle 1.

But angle 1 and angle 3 are alternate angles.

And if a line cuts two straight lines making the alternate angles equal, those lines are parallel — Theorem 23.

Therefore AB is parallel to CD. **Q.E.D.**

THEOREM 25

What parallels give back A THEOREM

AB and CD are two parallel straight lines, cut by the straight line EGHF, which meets AB at G and meets CD at H.

The angle marked 1 and the angle marked 3 are alternate angles.

The angle marked 2 and the angle marked 3 are the two interior angles on one side.

We are to prove two things: that angle 1 equals angle 3, and that angle 2 and angle 3 together equal two right angles.

Suppose angle 1 were greater than angle 3.

Add angle 2 to each. Then angle 1 plus angle 2 would be greater than angle 3 plus angle 2.

But angle 1 plus angle 2 equals two right angles, since they are adjacent angles on the straight line AB — Theorem 11.

So two right angles would be greater than angle 3 plus angle 2.

That is: the two interior angles on that side would together be less than two right angles.

But when that happens, the two lines meet on that side, if they are extended far enough — Postulate 5.

So AB and CD would meet. But they are given as parallel: they never meet, however far they are extended.

That cannot be. So angle 1 is not greater than angle 3.

By the very same argument, angle 1 is not less than angle 3.

Therefore angle 1 equals angle 3. That is the first thing.

Now, angle 2 plus angle 1 equals two right angles — Theorem 11.

And angle 1 equals angle 3.

Therefore angle 2 plus angle 3 equals two right angles. That is the second thing. **Q.E.D.**

THEOREM 26

Transitivity of parallelism A THEOREM

Let A, B and C be three straight lines, with A parallel to B, and B parallel to C.

We are to prove that A is parallel to C.

Cut all three of them with a fourth straight line D — Postulate 1 — making the angles marked 1, 2, 3 and 4.

Angle 1 equals angle 2, because A is parallel to B — Theorem 25.

Angle 2 equals angle 3, because they are vertical angles — Theorem 13.

Therefore angle 1 equals angle 3 — Common Notion 1.

Angle 3 equals angle 4, because B is parallel to C — Theorem 25.

Therefore angle 1 equals angle 4 — Common Notion 1.

But angle 1 and angle 4 are the alternate angles made by the line D cutting the lines A and C.

And when the alternate angles are equal, the two lines are parallel — Theorem 23.

Therefore A is parallel to C. **Q.E.D.**

THEOREM 27

Draw a parallel through a point A CONSTRUCTION

Given the straight line AB, and the point P, which is not on it.

We are to draw through P a straight line parallel to AB.

Take any point X on AB.

Join P to X — Postulate 1. This makes the angle PXB.

At P, make the angle LPX equal to the angle PXB — Theorem 20.

Now PL and AB are cut by the line PX, and the alternate angles LPX and PXB are equal.

When the alternate angles are equal, the two lines are parallel — Theorem 23.

Therefore PL is parallel to AB, and it passes through P. **Q.E.F.**

THEOREM 28

The angle sum of a triangle A THEOREM

Take any triangle ABC. Its three angles are marked 1 at C, 2 at A, and 3 at B.

Extend the side BC beyond C to a point X. This makes the exterior angle ACX.

We are to prove two things: that the exterior angle ACX equals angle 2 and angle 3 added together, and that angle 1, angle 2 and angle 3 added together equal two right angles.

Through C draw the line CP parallel to BA — Theorem 27.

This divides the exterior angle ACX into two parts: the angle marked 4 and the angle marked 5.

Angle 2 equals angle 4, because BA is parallel to CP — Theorem 25.

Angle 3 equals angle 5, because BA is parallel to CP — Theorem 25.

Add them together: angle 2 plus angle 3 equals angle 4 plus angle 5 — Common Notion 2.

But angle 4 plus angle 5 is the whole exterior angle ACX.

Therefore the exterior angle ACX equals angle 2 and angle 3 added together. That is the first thing.

Now add angle 1 to both sides: angle 1 plus angle 2 plus angle 3 equals angle 1 plus angle 4 plus angle 5 — Common Notion 2.

But angle 1, angle 4 and angle 5 are the angles standing at C on the straight line BCX, so together they equal two right angles — Theorem 11.

Therefore angle 1, angle 2 and angle 3 added together equal two right angles. **Q.E.D.**

THEOREM 29

The third angle is equal A THEOREM

Here are two triangles, ABC and DEF.

In the first, the angles are marked 1, 2 and 3. In the second, they are marked 4, 5 and 6.

We are given that angle 1 equals angle 4, and that angle 2 equals angle 5.

We are to prove that angle 3 equals angle 6.

In any triangle the three angles added together equal two right angles — Theorem 28.

So angle 1 plus angle 2 plus angle 3 equals two right angles.

And angle 4 plus angle 5 plus angle 6 equals two right angles.

In the first of these, put angle 4 in the place of angle 1, and angle 5 in the place of angle 2, since they are equal to them.

Then angle 4 plus angle 5 plus angle 3 equals two right angles — Common Notion 1.

So angle 4 plus angle 5 plus angle 3 equals angle 4 plus angle 5 plus angle 6 — Common Notion 1.

Take angle 4 and angle 5 away from each side. When equals are taken from equals, the remainders are equal — Common Notion 3.

Therefore angle 3 equals angle 6. **Q.E.D.**

THEOREM 30

One pair parallel and equal makes a parallelogram A THEOREM

Look at the quadrilateral ABCD.

In it, AB and CD are both parallel and equal.

We are to prove that it is a parallelogram — that AC and BD are parallel too, and that AC equals BD.

Join A to D — Postulate 1. This makes two triangles, ACD and ABD.

Since AB is parallel to CD, and AD cuts them both, the alternate angles are equal: the angle marked 1 equals the angle marked 2 — Theorem 25.

AB equals CD, which is given.

And AD belongs to both triangles alike.

So a side, the angle between, and the next side are equal, each to each.

Therefore triangle ACD and triangle ABD are equal in every corresponding part — Side-Angle-Side, Theorem 2.

Therefore the angle CAD equals the angle BDA.

But those are the alternate angles made by the line AD cutting AC and BD.

When the alternate angles are equal, the lines are parallel — Theorem 23.

So AC is parallel to BD.

And AC equals BD, these being corresponding sides of the two equal triangles.

Therefore ABCD is a parallelogram. **Q.E.D.**

THEOREM 31

The properties of a parallelogram A THEOREM

ABCD is a parallelogram: AB is parallel to CD, and BC is parallel to AD — Definition 23.

We are to prove three things: that its opposite sides are equal, that its opposite angles are equal, and that a diagonal cuts it into two equal halves.

Draw the diagonal AC — Postulate 1.

Since AB is parallel to CD, and AC cuts them, the alternate angles are equal: angle 1 equals angle 2 — Theorem 25.

Since BC is parallel to AD, and AC cuts them, the alternate angles are equal: angle 4 equals angle 3 — Theorem 25.

And AC belongs to both triangle ABC and triangle ADC alike.

So in the two triangles, two angles and the side joining them are equal to two angles and the side joining them.

Therefore the two triangles are equal in every corresponding part, and equal in area — Angle-Side-Angle, Theorem 21.

Since triangle ABC and triangle ADC are equal in area, the diagonal AC cuts the parallelogram into two equal halves.

Since they are equal in every corresponding part, AB equals CD, and BC equals DA.

And the angle marked 5 equals the angle marked 6; that is, the angle ABC equals the angle ADC.

Finally, angle 1 plus angle 4 equals angle 2 plus angle 3; that is, the angle BAD equals the angle BCD. **Q.E.D.**

THEOREM 32

Parallelograms on the same base and in the same parallels

A THEOREM

Here are two parallel lines, AF and BE.

Between them stand two parallelograms on the same base BE: the parallelogram ABED, and the parallelogram CBEF.

We are to prove that they have equal areas.

AD equals BE, being opposite sides of the parallelogram ABED — Theorem 31.

CF equals BE, being opposite sides of the parallelogram CBEF — Theorem 31.

Therefore AD equals CF — Common Notion 1.

Add CD to each of them. Then AC equals DF — Common Notion 2.

Now look at triangle ABC and triangle DEF.

AC equals DF, just proved.

AB equals DE, being opposite sides of the parallelogram ABED — Theorem 31.

BC equals EF, being opposite sides of the parallelogram CBEF — Theorem 31.

The three sides of the one are equal to the three sides of the other, each to each.

Therefore the two triangles are equal, and equal in area — Side-Side-Side, Theorem 6.

Now name the areas on the figure by their numbers.

Triangle ABC is area 1 together with area 4. Triangle DEF is area 4 together with area 3.

So area 1 plus area 4 equals area 4 plus area 3.

Take area 4 away from each side — Common Notion 3. Then area 1 equals area 3.

Add area 2 to each side — Common Notion 2. Then area 1 plus area 2 equals area 2 plus area 3.

But area 1 with area 2 is the parallelogram ABED, and area 2 with area 3 is the parallelogram CBEF.

Therefore the two parallelograms have equal areas. **Q.E.D.**

THEOREM 33

Triangles on the same base and in the same parallels A THEOREM

Two triangles, ABC and ABG, stand on the same base AB and between the same two parallels.

We are to prove that they have equal areas.

Through A draw AK parallel to BG, making the parallelogram ABGK — Theorem 27.

Through B draw BL parallel to AC, making the parallelogram ABLC — Theorem 27.

These two parallelograms stand on the same base AB and between the same parallels.

Therefore they have equal areas — Theorem 32.

Now, a diagonal cuts a parallelogram into two equal halves — Theorem 31.

So triangle ABG is half of the parallelogram ABGK, and triangle ABC is half of the parallelogram ABLC.

But the halves of equal things are equal.

Therefore triangle ABG and triangle ABC have equal areas.

And it follows that a triangle standing on the same base and between the same parallels as a parallelogram is half of that parallelogram. **Q.E.D.**

THEOREM 34

The complements about a diagonal A THEOREM

Take the parallelogram ABCD, and draw the diagonal AC.

Pick any point K on that diagonal.

Through K draw the line HKG parallel to AB, and the line EKF parallel to AD — Theorem 27.

This makes two small parallelograms sitting on the diagonal, and two others off the diagonal — the ones marked 1 and 2. Those two are called the complements.

We are to prove that the two complements are equal in area.

A diagonal cuts a parallelogram into two equal halves — Theorem 31.

So AK cuts the first small parallelogram in half: the area marked 5 equals the area marked 6.

And KC cuts the second small parallelogram in half: the area marked 3 equals the area marked 4.

Add these together: area 5 plus area 3 equals area 6 plus area 4 — Common Notion 2.

Now the whole diagonal AC cuts the whole parallelogram ABCD in half — Theorem 31.

So area 5 plus area 3 plus area 1 equals area 6 plus area 4 plus area 2.

Take the equal amounts away from each side — area 5 with area 3 from the one, area 6 with area 4 from the other — Common Notion 3.

Therefore area 1 equals area 2. The two complements are equal. **Q.E.D.**

THEOREM 35

How to make a square A CONSTRUCTION

Given the straight line AB.

We are to build a square on it, with AB as one of its sides.

From A draw AE perpendicular to AB, and make it as long as you need — Theorem 9.

On it cut off AD equal to AB, by drawing a circle about A with radius AB — Postulate 3.

Through D draw DK parallel to AB — Theorem 27.

Through B draw BM parallel to AD — Theorem 27.

Let these two lines meet at the point C.

Now ABCD has both pairs of opposite sides parallel, so it is a parallelogram — Definition 23.

The angle marked 1 and the angle marked 2 are the two interior angles on one side, where AD cuts the parallels AB and DC.

So angle 1 plus angle 2 equals two right angles — Theorem 25.

But angle 1 is a right angle, since we made AD perpendicular to AB.

Therefore angle 2 is a right angle also.

In a parallelogram the opposite angles are equal — Theorem 31.

So the angle marked 3, which is opposite angle 1, is a right angle; and the angle marked 4, which is opposite angle 2, is a right angle.

All four angles are right angles.

Again, in a parallelogram the opposite sides are equal — Theorem 31.

So DC equals AB, and CB equals AD.

But we made AD equal to AB.

Therefore AB, AD, DC and CB are all equal to one another.

ABCD has four equal sides and four right angles.

Therefore ABCD is a square — Definition 23. **Q.E.F.**

THEOREM 36

The Pythagorean Theorem A THEOREM

Triangle ABC has a right angle at A.

On each of its three sides a square has been built: the square ABFG on AB, the square ACKH on AC, and the square BCED on the hypotenuse BC — Theorem 35.

We are to prove that the square on the hypotenuse equals the other two squares added together.

From A drop AL perpendicular to DE — Theorem 10. This divides the square BCED into two rectangles: the rectangle BL and the rectangle CL.

Join A to D, and join C to F — Postulate 1.

First, notice that the angle GAB and the angle BAC are both right angles, so together they make two right angles. Therefore GAC is one straight line — Theorem 12.

Now look at triangle FBC and triangle ABD.

FB equals AB, because ABFG is a square.

BC equals BD, because BCED is a square.

And the angle FBC equals the angle ABD, because each of them is the angle ABC together with a right angle.

So a side, the angle between, and the next side are equal, each to each.

Therefore triangle FBC and triangle ABD are equal — Side-Angle-Side, Theorem 2.

Double them. The doubles of equal areas are equal, so twice triangle FBC equals twice triangle ABD — Common Notion 2.

Now, twice triangle FBC equals the square ABFG.

For that triangle and that square stand on the same base FB, and between the same parallels FB and GAC — and a triangle on the same base and in the same parallels as a parallelogram is half of it — Theorem 33.

And twice triangle ABD equals the rectangle BL.

For that triangle and that rectangle stand on the same base BD, and between the same parallels BD and AL — Theorem 33.

Since their doubles are equal, the square ABFG equals the rectangle BL.

By exactly the same argument on the other side, the square ACKH equals the rectangle CL.

Add these together: the square ABFG and the square ACKH together equal the rectangle BL and the rectangle CL together — Common Notion 2.

But the rectangle BL and the rectangle CL together are the whole square BCED.

Therefore the square on the hypotenuse equals the sum of the squares on the other two sides. **Q.E.D.**

THEOREM 37

Converse of the Pythagorean Theorem A THEOREM

In triangle ABC, the square on AB equals the square on AC and the square on BC added together.

We are to prove that the angle opposite AB — the angle ACB — is a right angle.

From C draw CN at right angles to AC — Theorem 9 — and cut off NC equal to BC.

Join N to A — Postulate 1.

Now triangle ACN has a right angle at C.

So by the Pythagorean Theorem — Theorem 36 — the square on AN equals the square on AC and the square on NC added together.

But NC equals BC, so the square on NC equals the square on BC.

Therefore the square on AN equals the square on AC and the square on BC added together.

And we were given that the square on AB equals the square on AC and the square on BC added together.

Things equal to the same thing are equal to each other — Common Notion 1.

Therefore the square on AN equals the square on AB, and so AN equals AB.

Now look at triangle ANC and triangle ABC.

AN equals AB, just proved.

NC equals BC, by the construction.

And CA belongs to both triangles alike.

The three sides of the one are equal to the three sides of the other, each to each.

Therefore their corresponding angles are equal — Side-Side-Side, Theorem 6.

Therefore the angle ACN equals the angle ACB.

But the angle ACN is a right angle, since we made it so.

Therefore the angle ACB is a right angle. **Q.E.D.**